



GATE ME 90 Days Challenge

Day 7/90 · Practice Set with Detailed Solutions

SUBJECT

Strength of Materials

TODAY'S TOPIC

SFD & BMD

COUNTDOWN

103 days to GATE

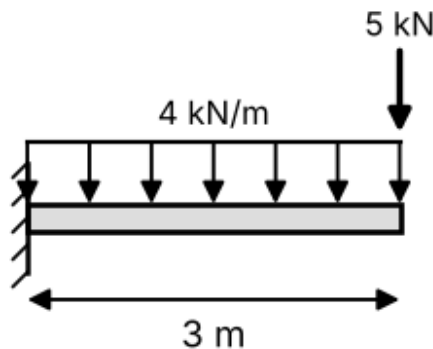
How to use this set: 15 questions (8 MCQ, 7 NAT) on shear force and bending moment diagrams for cantilevers, simply supported and overhanging beams under point loads, UDLs, varying loads and couples. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 (MCQ)

The 3 m cantilever shown carries a UDL of 4 kN/m over its whole length and a 5 kN point load at the free end. The bending moment at the fixed end has magnitude



(A) 33 kN·m

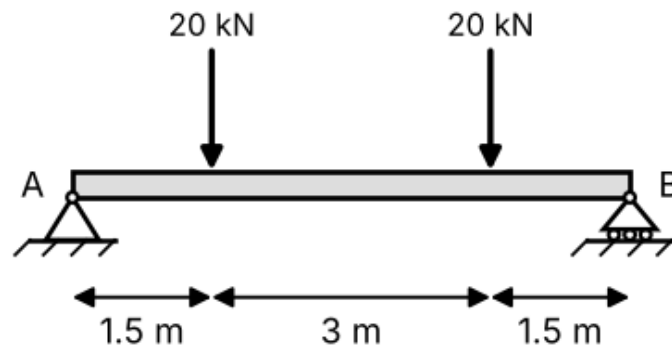
(B) 51 kN·m

(C) 24 kN·m

(D) 17 kN·m

Q2 (NAT)

The simply supported beam shown carries two 20 kN loads placed symmetrically. The maximum bending moment (in kN·m) is _____

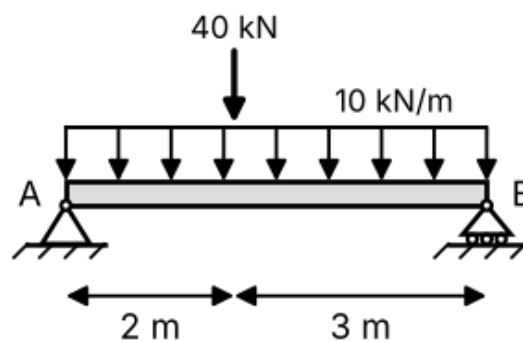
**Q3** (MCQ)

A simply supported beam of span L carries a UDL w over the whole span. The shear force at a section $L/4$ from a support has magnitude

- (A) $wL/2$ (B) $wL/4$
(C) $3wL/4$ (D) $wL/8$

Q4 (NAT)

The simply supported beam shown carries a UDL of 10 kN/m over its whole 5 m span and a 40 kN point load at 2 m from A. The maximum bending moment (in kN·m) is _____

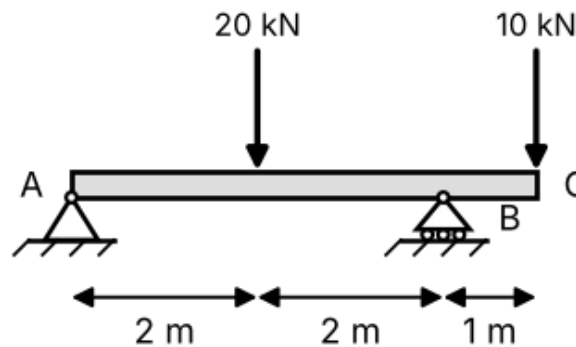
**Q5** (MCQ)

A sudden vertical jump in the bending moment diagram of a beam occurs at the section where

- (A) a point load is applied (B) a UDL starts
(C) a concentrated couple is applied (D) there is a support

Q6 (NAT)

The beam ABC shown has a span AB of 4 m and an overhang BC of 1 m. It carries 20 kN at the middle of AB and 10 kN at C. The distance of the point of contraflexure from A (in m) is _____

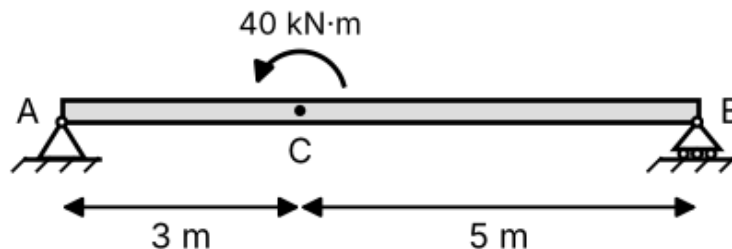
**Q7** (MCQ)

A 4 m cantilever carries a load varying linearly from zero at the free end to 6 kN/m at the fixed end. The bending moment at the fixed end has magnitude

- (A) 16 kN·m (B) 32 kN·m
(C) 48 kN·m (D) 24 kN·m

Q8 (NAT)

The simply supported beam AB shown (span 8 m) carries only an anticlockwise couple of 40 kN·m at C, 3 m from A. The maximum magnitude of the bending moment (in kN·m) is _____

**Q9** (MCQ)

The SFD of a simply supported beam of span 6 m is a single straight line falling from +30 kN at A to -30 kN at B. The maximum bending moment in the beam is

- (A) 90 kN·m (B) 30 kN·m
(C) 45 kN·m (D) 180 kN·m

Q10 (NAT)

A 3 m cantilever carries a UDL of 10 kN/m only over the 2 m length next to its free end (the 1 m next to the support is unloaded). The magnitude of the bending moment at the fixed end (in kN·m) is _____

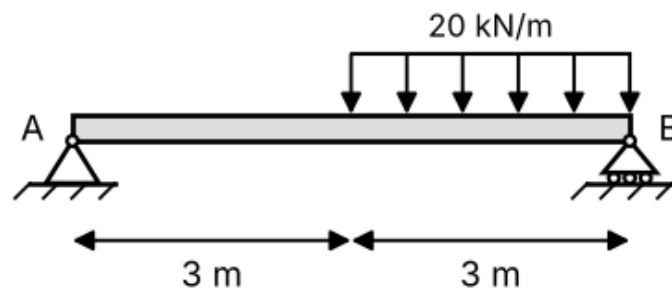
Q11 (MCQ)

A beam of length L rests on two supports placed symmetrically at a distance a from each end and carries a UDL over its whole length. For the least value of the maximum bending moment (hogging at the supports = sagging at mid-span), a should be about

- (A) $0.25L$ (B) $0.293L$
(C) $0.167L$ (D) $0.207L$

Q12 (NAT)

The simply supported beam shown (span 6 m) carries a UDL of 20 kN/m over its right half. The maximum bending moment (in $\text{kN}\cdot\text{m}$) is _____

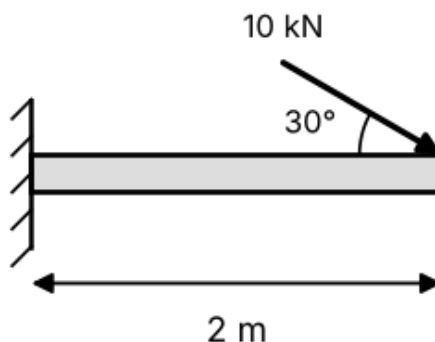
**Q13** (MCQ)

On a portion of a beam between two point loads, with no other load on that portion, the bending moment diagram is

- (A) a parabola (B) a straight line
(C) a cubic curve (D) a rectangle always

Q14 (NAT)

A 2 m cantilever carries a 10 kN load at its free end inclined at 30° to the axis of the beam, as shown. The magnitude of the bending moment at the fixed end (in $\text{kN}\cdot\text{m}$) is _____

**Q15** (MCQ)

A simply supported beam of span L carries two equal point loads W at $L/3$ and $2L/3$ from one support. The maximum bending moment is

- (A) $WL/3$ (B) $WL/4$
(C) $WL/6$ (D) $2WL/9$

Answer Key

Q	Type	Answer
Q1	MCQ	(A) 33 kN·m
Q2	NAT	30
Q3	MCQ	(B) $wL/4$
Q4	NAT	78
Q5	MCQ	(C) a concentrated couple is applied
Q6	NAT	3.2
Q7	MCQ	(A) 16 kN·m
Q8	NAT	25
Q9	MCQ	(C) 45 kN·m
Q10	NAT	40
Q11	MCQ	(D) 0.207L
Q12	NAT	50.63 (accept 50.3 to 51.0)
Q13	MCQ	(B) a straight line
Q14	NAT	10
Q15	MCQ	(A) $WL/3$

Detailed Solutions

Q1 Answer: (A) 33 kN·m

Concept: For a cantilever, take moments of all loads on the free side of the section; no reactions are needed.

From the point load: $5 \times 3 = 15 \text{ kN·m}$

From the UDL: $(4 \times 3) \times 1.5 = 18 \text{ kN·m}$

$M = 15 + 18 = \mathbf{33 \text{ kN·m}}$ (hogging); the SF at the fixed end is $5 + 12 = 17 \text{ kN}$.

Exam tip: 51 kN·m results from placing the UDL resultant at the free end instead of at its centroid.

Q2 Answer: 30

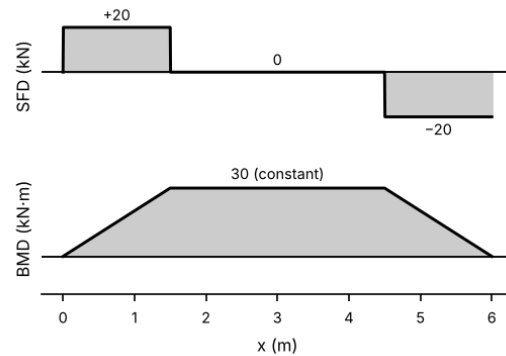
Concept: By symmetry each reaction is half the total load.

$$R_A = R_B = 20 \text{ kN}$$

Under each load: $M = 20 \times 1.5 = 30 \text{ kN}\cdot\text{m}$

Between the loads the SF is zero, so the BM stays constant at $30 \text{ kN}\cdot\text{m}$: this middle region is in **pure bending**, the set-up used in four-point bending tests.

Exam tip: Zero SF over a length means constant BM over that length.

**Q3 Answer: (B) $wL/4$**

Concept: $V(x) = R_A - wx$ with $R_A = wL/2$.

$$V(L/4) = wL/2 - wL/4 = wL/4$$

The SFD is a straight line from $+wL/2$ to $-wL/2$, crossing zero at midspan.

Exam tip: The BM there is $M = (wL/2)(L/4) - w(L/4)^2/2 = 3wL^2/32$.

Q4 Answer: 78

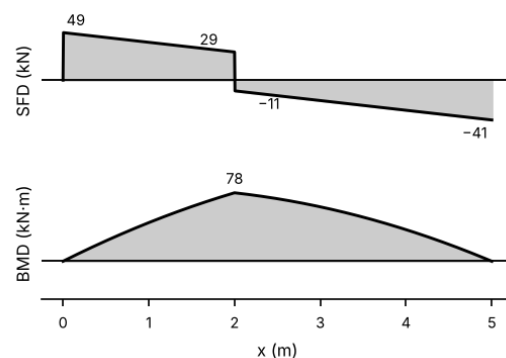
Concept: Find the reactions, then locate where the SF changes sign; the BM is maximum there.

$$R_A = 40 \times 3/5 + 50/2 = 24 + 25 = 49 \text{ kN}; R_B = 90 - 49 = 41 \text{ kN}$$

SF just left of the point load: $49 - 10 \times 2 = +29 \text{ kN}$; just right: $29 - 40 = -11 \text{ kN} \rightarrow$ sign change at $x = 2 \text{ m}$

$$M_{\max} = 49 \times 2 - 10 \times 2^2/2 = 98 - 20 = 78 \text{ kN}\cdot\text{m}$$

Exam tip: With a point load present, check whether the SF crosses zero inside a UDL segment or jumps through zero at the load. Here it jumps.



Q5 Answer: (C) a concentrated couple is applied

Concept: $dM/dx = V$. A point load makes V jump, so the BMD only changes slope there (a kink).

A concentrated couple M_0 adds M_0 to the moment over an infinitesimal length, so the **BMD jumps by M_0** while the SFD is unaffected.

A UDL changes the slope of the SFD; a support reaction is just a point force.

Exam tip: SFD jumps $\leftrightarrow$ point forces; BMD jumps $\leftrightarrow$ point couples.

Q6 Answer: 3.2

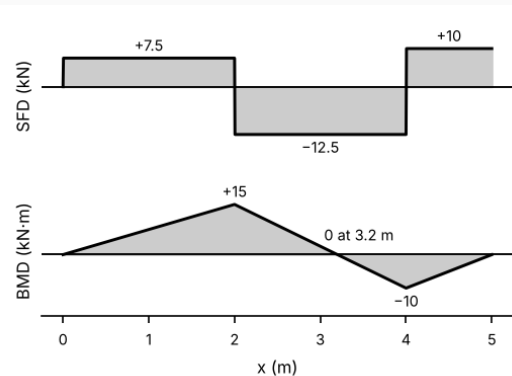
Concept: Point of contraflexure: $BM = 0$ (with a sign change) away from a free end or simple support.

Moments about B: $R_A \times 4 = 20 \times 2 - 10 \times 1 \rightarrow R_A = 7.5 \text{ kN}$; $R_B = 22.5 \text{ kN}$

For $2 \text{ m} < x < 4 \text{ m}$: $M = 7.5x - 20(x - 2) = 40 - 12.5x$

$M = 0 \rightarrow x = 3.2 \text{ m}$; M is $+15 \text{ kN}\cdot\text{m}$ under the 20 kN load and $-10 \text{ kN}\cdot\text{m}$ at B.

Exam tip: Overhanging beams almost always have a point of contraflexure; sketch the BMD sign at each key point first.

**Q7 Answer: (A) 16 kN·m**

Concept: Replace the triangular load by its resultant at the centroid of the triangle.

Resultant $= \frac{1}{2} \times 6 \times 4 = 12 \text{ kN}$, acting at $L/3 = 4/3 \text{ m}$ from the fixed end

$M = 12 \times 4/3 = 16 \text{ kN}\cdot\text{m} = wL^2/6$

Exam tip: If the triangle were reversed (maximum at the free end), the lever arm would be $2L/3$ and $M = wL^2/3 = 32 \text{ kN}\cdot\text{m}$.

Q8 Answer: 25

Concept: The reactions form a couple: $R = M_0/L$. Then the BMD is two parallel straight lines with a jump of M_0 at C.

$R = 40/8 = 5 \text{ kN}$; $R_A = 5 \text{ kN}$ upward, $R_B = 5 \text{ kN}$ downward

Just left of C: $M = 5 \times 3 = +15 \text{ kN}\cdot\text{m}$

An anticlockwise couple lowers the BM by 40: just right of C, $M = 15 - 40 = -25 \text{ kN}\cdot\text{m}$, then it rises linearly to 0 at B.

$|M|_{\max} = 25 \text{ kN}\cdot\text{m}$

Exam tip: The jump splits as M_0a/L and M_0b/L on either side: $40 \times 3/8 = 15$ and $40 \times 5/8 = 25$.

Q9 Answer: (C) 45 kN·m**Concept:** $dV/dx = -w$ and the area of the SFD gives the change in BM.Slope = $60/6 = 10 \text{ kN/m} \rightarrow$ a UDL of 10 kN/m over the whole span $M_{\max} =$ area of the positive SFD triangle = $\frac{1}{2} \times 3 \times 30 = \mathbf{45 \text{ kN}\cdot\text{m}}$ ($= wL^2/8$ ✓)**Exam tip:** Reading loads from an SFD: slope = load intensity, jumps = point loads.**Q10 Answer: 40****Concept:** Resultant of a partial UDL acts at the middle of the loaded length.Resultant = $10 \times 2 = 20 \text{ kN}$, at 1 m from the free end, i.e. 2 m from the support $M = 20 \times 2 = \mathbf{40 \text{ kN}\cdot\text{m}}$ Over the unloaded 1 m the SF is constant (20 kN) and the BM varies linearly.**Exam tip:** If the same UDL were next to the support instead, M would be only $20 \times 1 = 20 \text{ kN}\cdot\text{m}$.**Q11 Answer: (D) 0.207L****Concept:** Equate the hogging moment over a support to the sagging moment at mid-span.Hogging: $wa^2/2$. Sagging at centre: $(wL/2)(L/2 - a) - wL^2/8 = wL^2/8 - wLa/2$ $wa^2/2 = wL^2/8 - wLa/2 \rightarrow 4a^2 + 4aL - L^2 = 0$ $a = L(\sqrt{2} - 1)/2 = \mathbf{0.207L}$ **Exam tip:** A classic GATE result; the supports are then $0.586L$ apart.**Q12 Answer: 50.63 (accept 50.3 to 51.0)****Concept:** The maximum BM is where the SF is zero, which lies inside the loaded half. $R_A = 60 \times 1.5/6 = 15 \text{ kN}$; $R_B = 45 \text{ kN}$ For $x > 3 \text{ m}$: $V = 15 - 20(x - 3) = 0 \rightarrow x = 3.75 \text{ m}$ $M = 15 \times 3.75 - 20 \times 0.75^2/2 = 56.25 - 5.63 = \mathbf{50.63 \text{ kN}\cdot\text{m}}$ **Exam tip:** The maximum is NOT at mid-span (M there is $45 \text{ kN}\cdot\text{m}$) and not under the centre of the load.**Q13 Answer: (B) a straight line****Concept:** $w = 0 \rightarrow V$ is constant $\rightarrow dM/dx$ is constant.So the BMD is a **straight line** (horizontal if $V = 0$ there, as in Q2).

A parabola needs a UDL; a cubic needs a linearly varying load.

Exam tip: Degree rule: load degree $n \rightarrow$ SF degree $n + 1 \rightarrow$ BM degree $n + 2$ (point loads count as 'no distributed load').**Q14 Answer: 10****Concept:** Only the component of the load perpendicular to the beam axis bends the beam; the axial component causes direct (axial) force.Transverse component = $10 \sin 30^\circ = 5 \text{ kN}$; axial component = $10 \cos 30^\circ = 8.66 \text{ kN}$ $M = 5 \times 2 = \mathbf{10 \text{ kN}\cdot\text{m}}$ **Exam tip:** $17.32 \text{ kN}\cdot\text{m}$ (using $\cos 30^\circ$) is the trap: check which angle is given, with the axis or with the vertical.

Q15 Answer: (A) $WL/3$

Concept: Symmetric loading: each reaction = W .

Under each load: $M = W \times L/3 = \mathbf{WL/3}$; between the loads the SF is zero and the BM stays at $WL/3$.

Compare: a single central load $2W$ would give $2W \times L/4 = WL/2$.

Exam tip: For loads at distance a from each support, $M_{\max} = Wa$ over the whole middle region.

What next?

Day 7 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124



Website test series



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Telegram: daily PDFs

Next, Day 8/90: Strength of Materials, Bending & Shear Stresses. Comment **GATE90** on our reel to get all links.

Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.